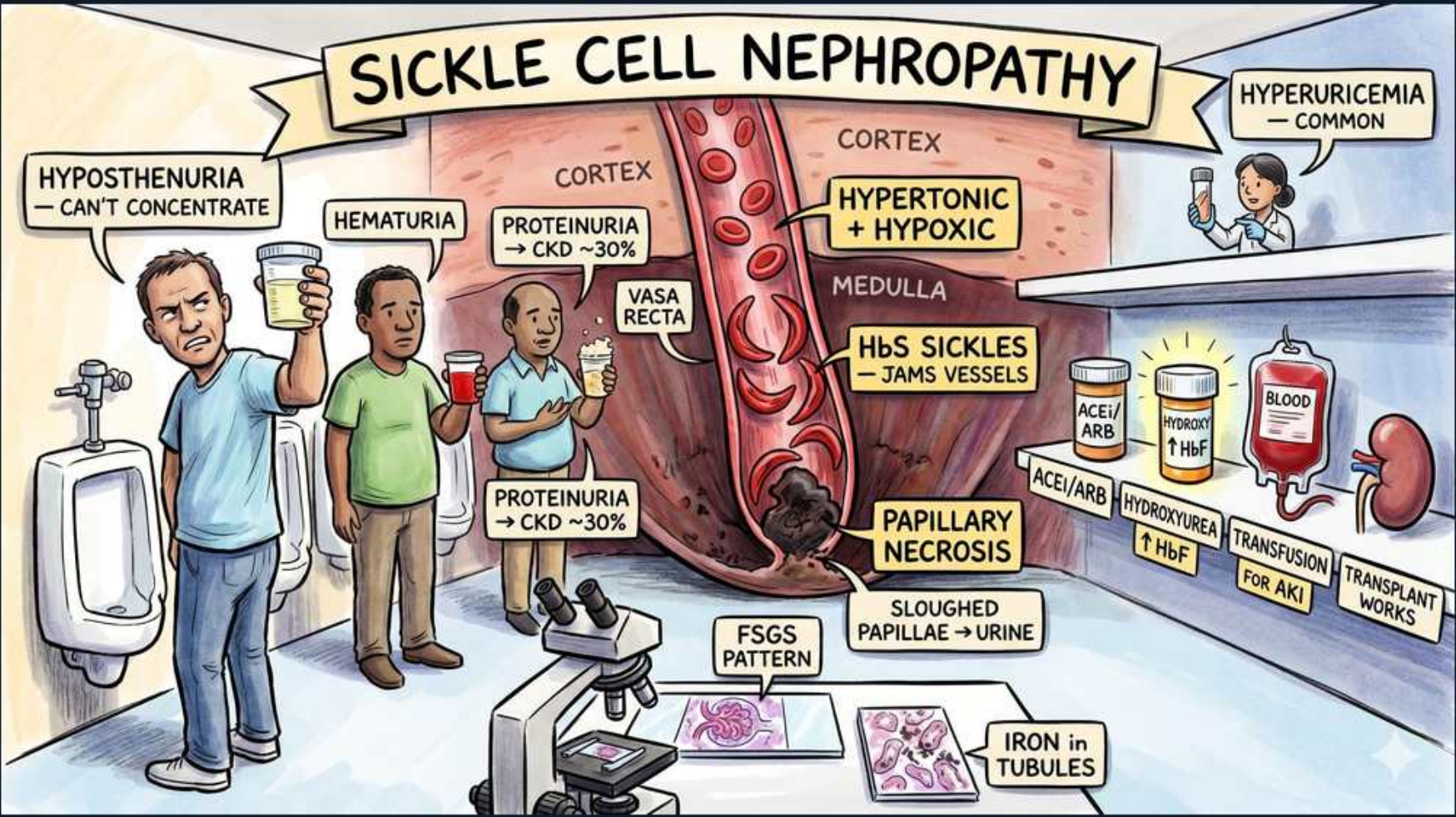


Glomerular (Nephrotic) — Sickle Cell Nephropathy



1. Sickle cell nephropathy banner

WHAT YOU SEE

Top banner SICKLE CELL NEPHROPATHY

WHAT IT ENCODES

Sickle cell nephropathy is the spectrum of renal injury caused by repeated RBC sickling within the renal medulla. Manifestations range from the earliest concentrating defect (hyposthenuria) and painless hematuria to papillary necrosis, glomerular FSGS with proteinuria, and ultimately CKD/ESKD. It occurs across the genotype spectrum — full disease (HbSS), HbSC, and even sickle TRAIT (HbAS) cause renal abnormalities because the medulla provokes sickling even at low HbS percentages.

BOARD TRAP

WRONG: 'Only patients with homozygous sickle cell disease (HbSS) get renal manifestations.' Discriminator — sickle TRAIT (HbAS), classically considered benign, still causes hematuria, hyposthenuria, papillary necrosis, and (rarely) renal medullary carcinoma, because the hypertonic/hypoxic medulla drives sickling even with a minority of HbS.

2. Hypertonic + hypoxic medulla

WHAT YOU SEE

Right of vessel 'HYPERTONIC + HYPOXIC' over medulla

WHAT IT ENCODES

The renal medulla is the perfect storm for HbS polymerization: it is hypertonic (high osmolality from the countercurrent gradient), acidic, and hypoxic (low pO2 from sluggish vasa recta flow). Each of these — low O2, low pH, high osmolality/dehydration — shifts the oxygen–hemoglobin relationship and promotes deoxygenated HbS polymerization → sickling. This is why the medulla, not the well-oxygenated cortex, bears the brunt of injury.

BOARD TRAP

WRONG: 'Sickling is worst in the renal CORTEX where filtration occurs.' Discriminator — sickling concentrates in the MEDULLA/vasa recta because of its hypertonic, acidic, hypoxic microenvironment; the cortex is relatively spared. Hydration and avoiding the hyperosmolar/hypoxic state mitigate medullary sickling.

3. HbS sickles / jams vessels

WHAT YOU SEE

Mid panel 'HbS SICKLES — jams vessels' along vasa recta

WHAT IT ENCODES

Deoxygenated HbS polymerizes into rigid fibers that deform RBCs into the sickle shape; these inflexible cells occlude the thin-walled vasa recta supplying the medulla → microvascular vaso-occlusion and ischemia. Repeated cycles cause the downstream papillary necrosis and microinfarcts. This vaso-occlusive ischemia is the central pathophysiologic mechanism uniting all the renal findings.

BOARD TRAP

WRONG: 'Direct immune-complex deposition in glomeruli is the primary mechanism of sickle nephropathy.' Discriminator — the core mechanism is mechanical VASO-OCCLUSION of the vasa recta by sickled cells causing medullary ischemia; glomerular injury (FSGS) is a SECONDARY hyperfiltration consequence, not immune-complex disease.

4. Papillary necrosis

WHAT YOU SEE

Dark necrotic tip 'PAPILLARY NECROSIS'

WHAT IT ENCODES

Medullary/vasa recta vaso-occlusion causes ischemic RENAL PAPILLARY NECROSIS — a classic, board-favorite complication. Presents with gross hematuria, flank/colicky pain, and sometimes a 'ring sign' (sloughed papilla outlined by contrast) on imaging. The mnemonic for papillary necrosis causes is POSTCARDS: Pyelonephritis, Obstruction, Sickle cell, TB, Cirrhosis/Chronic alcoholism, Analgesics (NSAIDs/phenacetin), Renal vein thrombosis, Diabetes mellitus, Systemic vasculitis.

BOARD TRAP

WRONG: 'Papillary necrosis in a hematuric patient is essentially pathognomonic for sickle cell disease.' Discriminator — papillary necrosis has a broad differential (POSTCARDS): the big three on boards are diabetes, analgesic nephropathy (NSAIDs/phenacetin), and sickle cell; also obstruction and pyelonephritis. Don't anchor on sickle alone.

5. Sloughed papillae → urine

WHAT YOU SEE

Note 'sloughed papillae → urine'

WHAT IT ENCODES

Necrotic papillae detach and pass into the collecting system/urine, producing gross hematuria, ureteral colic, and potential obstruction. On CT urography/IVP a sloughed papilla leaves a filling defect or calyceal 'ring sign,' which can be mistaken for a stone (radiolucent) or urothelial tumor.

BOARD TRAP

WRONG: 'A filling defect on urography in a sickle cell patient must be a urinary stone or tumor and warrants resection.' Discriminator — a calyceal filling defect from a SLOUGHED PAPILLA mimics stone/tumor but is managed supportively (hydration, pain control); recognizing papillary necrosis avoids unnecessary intervention.

6. Hyposthenuria

WHAT YOU SEE

Left figure at urinal 'HYPOSTHENURIA — can't concentrate'

WHAT IT ENCODES

Vasa recta damage disrupts the countercurrent multiplier, impairing urinary CONCENTRATING ability → hyposthenuria (dilute, fixed low-osmolality urine), nocturia, polyuria, and increased dehydration risk (which itself worsens sickling — a vicious cycle). It is the EARLIEST and nearly universal renal abnormality and, importantly, is present even in sickle TRAIT.

BOARD TRAP

WRONG: 'Hyposthenuria is a late finding seen only in advanced sickle cell disease.' Discriminator — hyposthenuria (impaired concentration) is the EARLIEST, near-universal defect, appearing even in sickle trait and young children; it is partially reversible early but becomes fixed with chronic medullary damage.

7. Hematuria

WHAT YOU SEE

Figure holding red urine cup 'HEMATURIA'

WHAT IT ENCODES

Painless gross or microscopic HEMATURIA arises from medullary microinfarcts and capillary congestion; it characteristically arises more often from the LEFT kidney (the longer left renal vein is compressed between the aorta and SMA — nutcracker-type effect). Hematuria occurs in both disease and trait. Always exclude renal medullary carcinoma in a young sickle-trait patient with painless hematuria.

BOARD TRAP

WRONG: 'Painless hematuria in a young Black patient with sickle trait is benign and needs no workup.' Discriminator — sickle trait is associated with RENAL MEDULLARY CARCINOMA, a rare highly aggressive tumor of young patients; unexplained painless hematuria warrants imaging to exclude it rather than reflexive reassurance.

8. Proteinuria → CKD ~30%

WHAT YOU SEE

Lower-left figure 'PROTEINURIA → CKD ~30%'

WHAT IT ENCODES

Chronic anemia and loss of nephrons drive compensatory glomerular HYPERFILTRATION (elevated single-nephron GFR), which over time causes albuminuria/proteinuria and glomerulosclerosis. Microalbuminuria is an early sentinel; roughly 30% of SCD patients progress to CKD, and sickle nephropathy is a leading cause of mortality in adults with SCD. Proteinuria magnitude predicts progression.

BOARD TRAP

WRONG: 'Proteinuria in sickle cell disease is purely tubular (from medullary damage) and does not signal progression.' Discriminator — the clinically important proteinuria is GLOMERULAR (hyperfiltration-driven albuminuria) and is a marker of progressive CKD; it is the target of ACEi/ARB therapy, unlike the benign concentrating/tubular defects.

9. FSGS pattern

WHAT YOU SEE

Microscope slide 'FSGS PATTERN'

WHAT IT ENCODES

The dominant glomerular lesion is SECONDARY focal segmental glomerulosclerosis (FSGS) from hyperfiltration injury; a membranoproliferative-like pattern and the aggressive COLLAPSING variant of FSGS are also described. Because it is secondary (adaptive/hyperfiltration), it responds to RAAS blockade and hyperfiltration control rather than to steroids.

BOARD TRAP

WRONG: 'The FSGS in sickle cell nephropathy is primary podocytopathy and should be treated with high-dose corticosteroids.' Discriminator — this is SECONDARY (hyperfiltration) FSGS; management is ACEi/ARB and treating the underlying disease, NOT immunosuppression. Steroids are reserved for primary FSGS, which has diffuse foot-process effacement and abrupt nephrotic syndrome.

10. Iron in tubules

WHAT YOU SEE

Bottom-right tissue 'IRON in TUBULES'

WHAT IT ENCODES

Chronic intravascular HEMOLYSIS releases free hemoglobin that is filtered and reabsorbed by proximal tubular cells, depositing hemosiderin → iron staining of the proximal tubules (and, with transfusions, systemic iron overload). This reflects the hemolytic phenotype of SCD and is distinct from the vaso-occlusive/glomerular injury pathways.

BOARD TRAP

WRONG: 'Tubular iron/hemosiderin deposition reflects glomerular immune-complex injury.' Discriminator — tubular hemosiderin comes from chronic INTRAVASCULAR HEMOLYSIS (free Hb reabsorbed by the proximal tubule), a separate mechanism from the glomerular hyperfiltration injury; it parallels other hemolytic markers (low haptoglobin, high LDH, indirect bilirubin).

11. Hyperuricemia common

WHAT YOU SEE

Top-right 'HYPERURICEMIA — common'

WHAT IT ENCODES

Hyperuricemia is common in SCD from two converging mechanisms: increased urate production from high marrow cell turnover (hemolysis) and decreased renal urate secretion as tubular function declines. This predisposes to gout and uric-acid nephrolithiasis; manage with hydration and urate-lowering therapy (allopurinol) when symptomatic.

BOARD TRAP

WRONG: 'Hyperuricemia in SCD is solely from overproduction, so reduced renal clearance can be ignored.' Discriminator — both mechanisms contribute: overproduction (hemolytic turnover) AND impaired tubular urate secretion as nephropathy advances. Attributing hyperuricemia to a single cause underestimates the additive role of declining renal function.

12. ACEi/ARB for proteinuria

WHAT YOU SEE

Pill bottles 'ACEi/ARB'

WHAT IT ENCODES

ACE inhibitors / ARBs are first-line for sickle-cell proteinuria/albuminuria: they lower intraglomerular pressure by dilating the efferent arteriole, reducing hyperfiltration, proteinuria, and CKD progression. Start when microalbuminuria/proteinuria appears; monitor creatinine and potassium after initiation (expect a small reversible eGFR dip).

BOARD TRAP

WRONG: 'Diuretics are the first-line agents to reduce proteinuria in sickle cell nephropathy.' Discriminator — the antiproteinuric/renoprotective agents are ACEi/ARB (efferent arteriolar dilation cuts hyperfiltration); aggressive diuresis is actually risky because volume depletion WORSENS medullary sickling. Treat the glomerular/hyperfiltration arm with RAAS blockade.

13. Hydroxyurea ↑HbF

WHAT YOU SEE

Bottle 'HYDROXYUREA ↑HbF'

WHAT IT ENCODES

Hydroxyurea (typical start ~15 mg/kg/day, titrated to effect/tolerance) raises fetal hemoglobin (HbF), which dilutes HbS and inhibits its polymerization, thereby reducing sickling, vaso-occlusive crises, and overall disease burden — a disease-modifying therapy that indirectly protects the kidney. Monitor for myelosuppression (its main dose-limiting toxicity).

BOARD TRAP

WRONG: 'Hydroxyurea works by directly blocking glomerular hyperfiltration in the kidney.' Discriminator — hydroxyurea is a DISEASE-MODIFYING agent that raises HbF and reduces sickling systemically; its renal benefit is indirect (less vaso-occlusion). The DIRECT antiproteinuric renal drug is the ACEi/ARB.

14. Transfusion / transplant

WHAT YOU SEE

Blood bag 'TRANSFUSION for AKI' and 'TRANSPLANT works'

WHAT IT ENCODES

Exchange or simple TRANSFUSION lowers the HbS fraction during severe events/AKI to interrupt vaso-occlusion. For ESKD, kidney TRANSPLANTATION offers better survival than chronic dialysis in SCD patients and is an effective option. Chronic transfusion carries iron-overload risk (manage with chelation).

BOARD TRAP

WRONG: 'Kidney transplantation is contraindicated in sickle cell disease and these patients should stay on dialysis.' Discriminator — transplantation is effective and improves survival over dialysis in SCD-ESKD; it is not contraindicated. Transfusion (often exchange) is the acute tool to drop HbS during severe vaso-occlusive/AKI events.
